

Benchmarking Information Extraction and Question Answering Pipelines from Resumes using Large Language Models and Layout-Aware Systems

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1 Introduction

Extracting structured information and answering questions from resumes is a high-stakes task with applications in university admissions and hiring. With the advancement of large language models (LLMs), retrieval-augmented generation (RAG), and layout-aware transformers, this problem can be tackled using a variety of pipelines. In this report, we benchmark four distinct approaches and evaluate their effectiveness using a common resume dataset annotated with structured fields and question-answering (QA) pairs.

2 Research Questions

1. **RQ1:** Which pipeline architecture is most effective for structured information extraction from resumes?
2. **RQ2:** How do layout-aware models compare with LLM-based pipelines in understanding spatially formatted resumes?
3. **RQ3:** What is the impact of retrieval strategy (1-step vs 2-step vs RAG) on answer accuracy and latency in resume QA tasks?
4. **RQ4:** Can LLM-based systems produce faithful rationales for their answers, and how reliable are their confidence scores?
5. **RQ5:** How does model size influence performance across retrieval and layout-based pipelines?

3 Pipelines and Model Selection

We benchmark the following four approaches:

3.1 1-Step Retrieval + LLM

The full resume and the question are passed to the LLM, which generates an answer directly.

- Models: Mistral Instruct 7B, Qwen 2B (find more later)

3.2 2-Step Retrieval + LLM

The relevant section of the resume is retrieved first, then passed to the LLM for answer generation.

- Models: Mistral Instruct 7B, Qwen 2B (more later)

3.3 RAG Systems

We evaluate state-of-the-art retrieval-augmented generation models:

- Atlas 11B
- Raven 11B
- SelfRAG-7B, SelfRAG-13B
- RECOMP 20B
- LLaMA Instruct 8B
- ChatQA-1.5 8B, ChatQA-1.5 70B

3.4 Layout-Aware Models

Models that consider both textual and layout information of documents:

- LayoutLMv3
- Donut
- FormNet
- LiLT

4 Experiments

4.1 Experiment 1: Structured Field Extraction

Goal: Evaluate F1 score for key fields (e.g., GPA, GRE, Publications).

RQ Addressed: RQ1, RQ2

4.2 Experiment 2: Resume QA Accuracy

Goal: Answer predefined questions like "What is the applicant's background in NLP?"

Metrics: QA accuracy

RQ Addressed: RQ1, RQ2, RQ3

4.3 Experiment 3: Retrieval Strategy Impact

Goal: Compare accuracy and latency between 1-step, 2-step, and RAG pipelines.

Metrics: Answer accuracy, average response time

RQ Addressed: RQ3

4.4 Experiment 4: Explainability and Calibration

Goal: Measure rationale correctness and model confidence alignment

Metrics: Rationale accuracy, Expected Calibration Error (ECE)

RQ Addressed: RQ4

4.5 Experiment 5: Model Size vs. Performance

Goal: Evaluate impact of model size in RAG pipeline (e.g., 7B vs 70B)

Metrics: Accuracy, Latency

RQ Addressed: RQ5